

05: From User Research to Conceptual Design

IN3065/INM355 | 2025–2026

User-Centred System Design

Dr Tracey Booth (she/her)

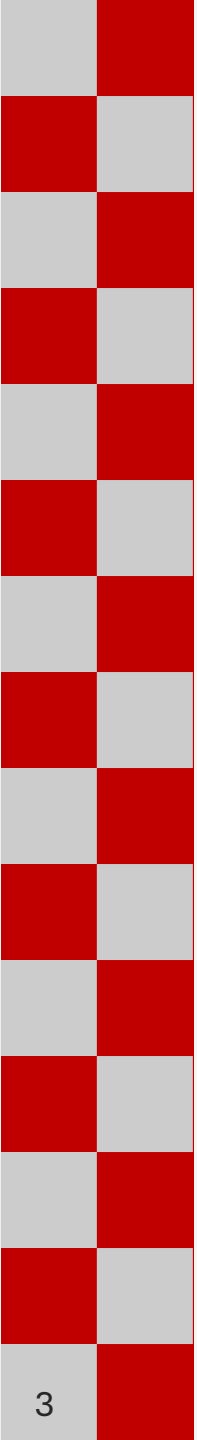
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Session 5: **Learning objectives**



By completing this session, including any practical activity and assigned readings, you should be able to:

- Define design goals and requirements, distilled from user research data, using techniques such as Point of View statements and user stories.
- Understand the role and value of conceptual design activities within a user-centred design process.
- Use ideation techniques to generate design ideas.
- Explore and document design ideas, using sketching



Progressing from User Research

Defining what we will design

Using the data from User Research



User research tells you about people, what they do, where they do it, etc.

What then?

- You need to define the design goal(s) or challenge that you will address, thinking from the user's perspective.
- Sometimes you will have a high-level statement of this from the outset, but may need to make it more specific
- We also need to make sure that our personas are tailored for the project—recall, personas aren't just summaries of our research data, they are framed for the challenge(s) we intend to address.

You might think about **problems**



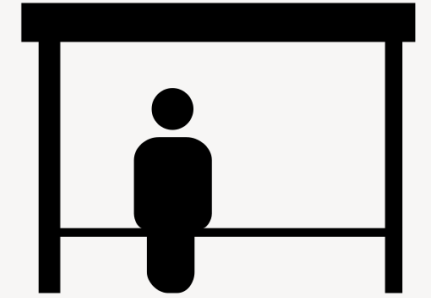
For example:

- Mrs Morton struggles to open pillboxes.
- Mrs Morton takes out-of-date medicines, believing them to be safe.
- Pet owners have to re-clean some areas more often, due to constant fur shedding or mess made during feeding.
- International tourists can feel unsure of local rules for behaviour in London Parks

Remember that **you**, not the users, will think about problems, based on what you have discovered through your user research.

[You are the designer.](#)

You might think about **opportunities**



For example:

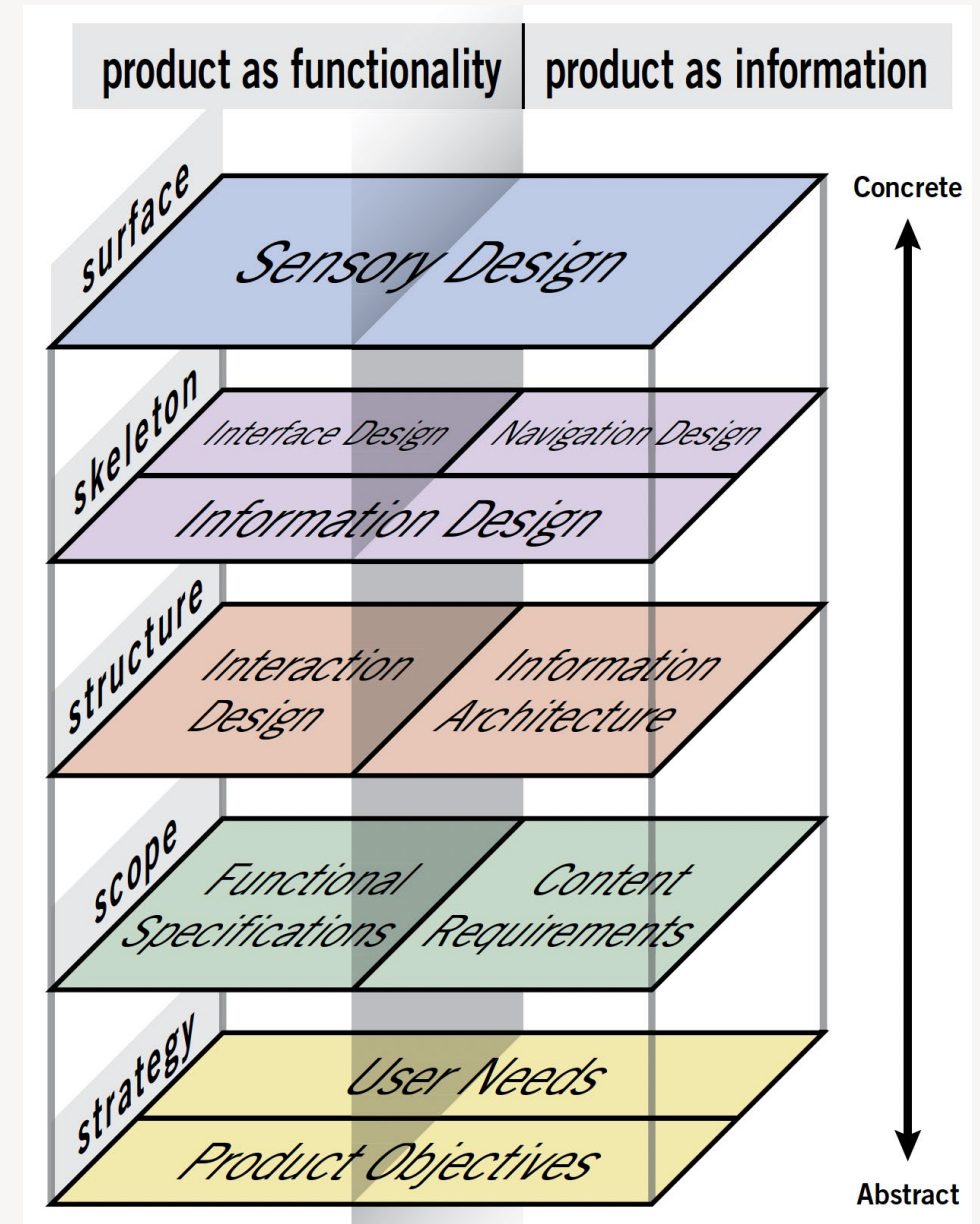
- To create a digital technology that turn routine commuting into a more enjoyable experience.
- To create a digital technology that will enable people to automate the cleaning of their floors to happen at convenient times.
- To create a digital technology that will enrich the experience that visitors have of a particular London park.

Remember that **you**, not the users, will think about opportunities, based on what you have discovered through user research.

[You are the designer.](#)

Strategy & scope

- We can use user research to empathise with users and identify their needs/goals.
- **Defining** the problem* *sharpens* the strategy, and begins to outline **scope**
 - What are we addressing?
 - For whom?
 - What is required?
- From abstract to more concrete, including articulating the goals for design.



*not always something that is *problematic*, but rather the *focus for design*

Image from JJ Garrett (2011)
The Elements of User Experience



Defining what we will design

Point of View (PoV) statements

/ user need statements

/ problem statements

Design goals/challenges articulated from the perspective of user need and value

Point of View (PoV) statement

You can capture these challenges in a Point of View statement:

[User] [Need] [Insight/Goal]

For example, user research looking for design opportunities to improve medication taking for older people might result in POV statements such as:

Mrs Morton

**needs support remembering to take her medication at the right times,
so that she can manage her health conditions safely and consistently.**

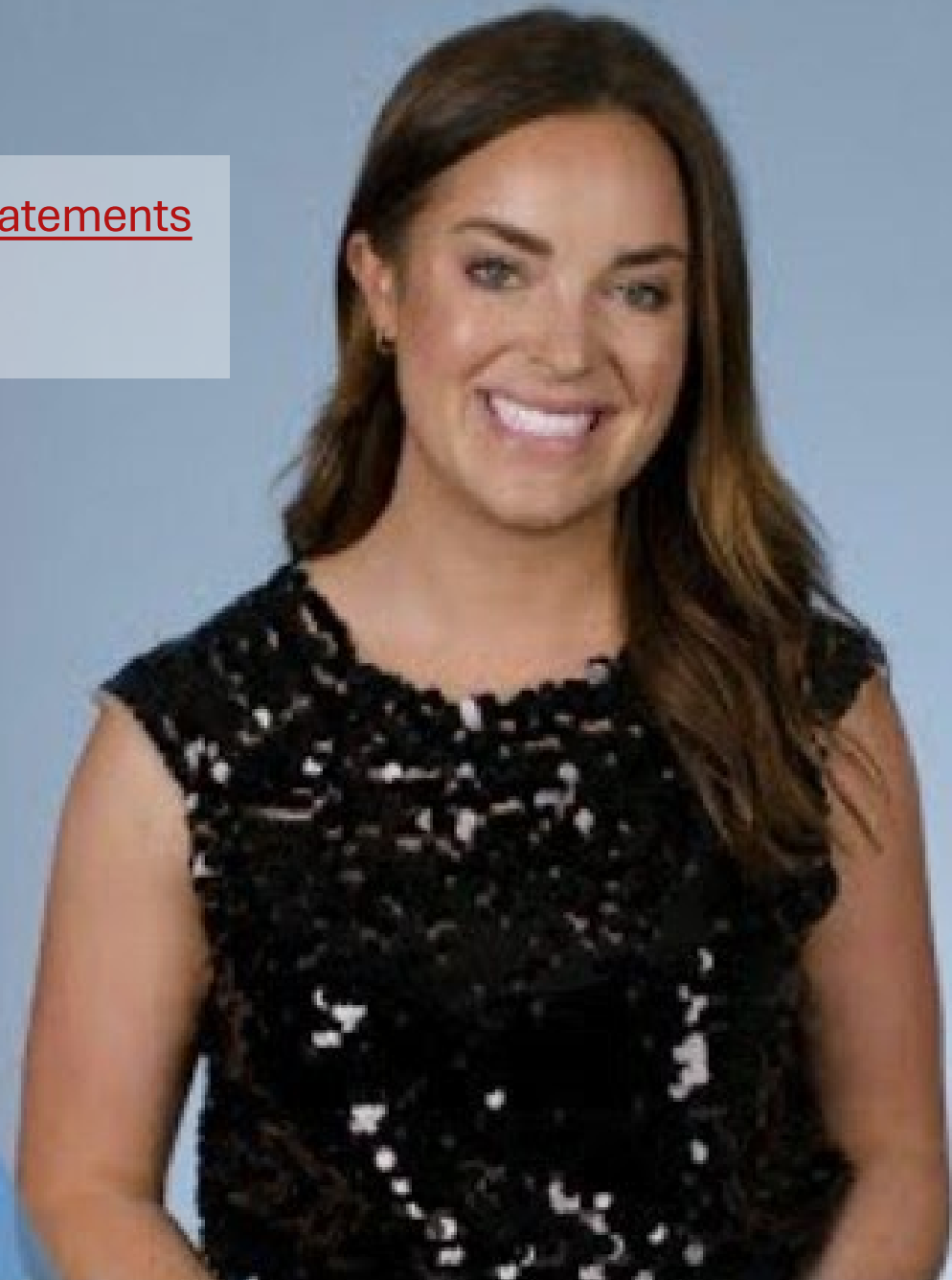
Note that the users may not have *articulated* a need. You may have identified a pain point or an opportunity to do something new/different.

NN/g

<https://www.nngroup.com/articles/user-need-statements>

<https://youtu.be/kT0ZqwdPYRM>

User Need Statements



Point of View **template**

User	Need	Insight/Goal

Adapted from: <https://www.interaction-design.org/literature/article/define-and-frame-your-design-challenge-by-creating-your-point-of-view-and-ask-how-might-we>

Exercise: Point of View statement

In your own time, practise writing one or more Point of View Statement for Mrs Morton or for your own persona.

User	Need	Insight/Goal

Articulating **design goals/challenges**

POV statements / User Need statements encapsulate what the design should achieve for a specific user—not what the system will contain.

They can also be applied to varying scopes (e.g. ‘parent’/‘umbrella’ vs ‘child’*)

They can help you articulate the design goal(s)/challenge(s) that your project will address.

They are:

- ✓ Grounded in research findings
- ✓ Shaped by project context
- ✓ Non-solution-specific
- ✓ Focused on outcomes

They are not:

- ✗ A persona
- ✗ A feature
- ✗ A screen...

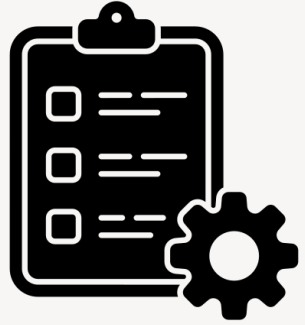
*see <https://www.nngroup.com/articles/user-need-statements>



Defining what we will design

Requirements

Requirements



- Requirements are statements that specify what a product must do and the conditions it must meet.
- They translate user needs and project goals into clear, testable expectations of:
 - **Behaviour** (what it must do)
 - **Quality** (how well it must do it)
 - **Constraints** (limits it must operate within)
- They can come from many sources: user research, stakeholders, business goals, domain experts, standards/regulation, and competitor/market analysis.
- Compared with POV/user-need statements, requirements are usually more system-centred than user-focused.

Some **types** of requirements

- **Functional requirements**

What the product must do (capabilities / features)

- **Non-functional requirements**

How well it must do it (quality attributes and constraints) e.g:

- Performance
- Usability & UX
- Security & privacy
- Reliability and safety
- Maintainability and portability, etc...

Example: vending machine

- A **functional** requirement might be that the machine must allow the customer to purchase a snack.
- A **non-functional** requirement (performance) might be that the machine has to process the transaction in, at the most, 10 seconds.
- A **non-functional** requirement (usability) might be an ease-of-use requirement: that 98% of users should be able to select their chosen item in less than 5 seconds.



Atomic requirement template (Volere)

Requirement No:

Requirement type:

Description: A one sentence statement of the intention of the requirement

Rationale: A justification of the requirement

Source: Who raised this requirement

Fit criterion: A measurement of the requirement such that it is possible to test if the solution matches the original requirement

Customer satisfaction:

Customer dissatisfaction:

Dependencies: A list of other requirements that have some dependency on this one.

Conflicts: Other requirements that cannot be implemented if this one is.

Supporting materials: Pointers to documents that illustrate and explain this requirement

History: Creation, changes, deletions etc.

Do not use this technique for your coursework!

Requirements as part of the **CW2 coursework**

- A small set of **research-informed** statements
- Because of the strict word limit, you will need to **prioritise key requirements** in the main report body.
 - Focus on what **must** be true (outcomes + key constraints).
 - If you do wish to include more, e.g. just for information, create a specific appendix.
- Each requirement should be **traceable** to a finding / your research.



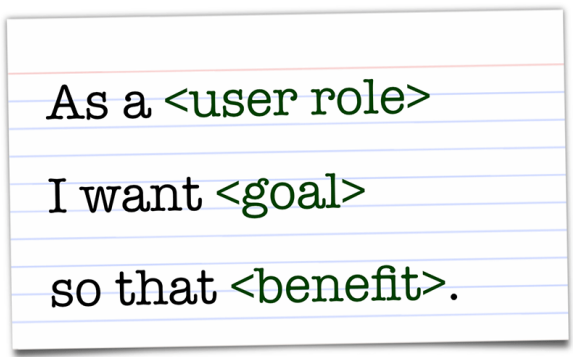


Defining what we will design

User Stories

User Stories

- From Agile software development
- Written from a user/customer perspective
 - “a short, simple description of a feature told **from the perspective of the person who desires the new capability**, usually a user or customer of the system.” (Mike Cohn)*
- As a [user type] I would like [some action/goal] so that [some reason/benefit]
- “Three Cs”: Card, Conversation, Confirmation (Jeffries, 2001)
- A lightweight placeholder to use in discussion at the appropriate time
 - (instead of detailed upfront specification)
- Usually stored/managed in a product backlog.
- See also <https://www.mountaingoatsoftware.com/agile/user-stories>



As a <user role>
I want <goal>
so that <benefit>.

User Stories in **design**

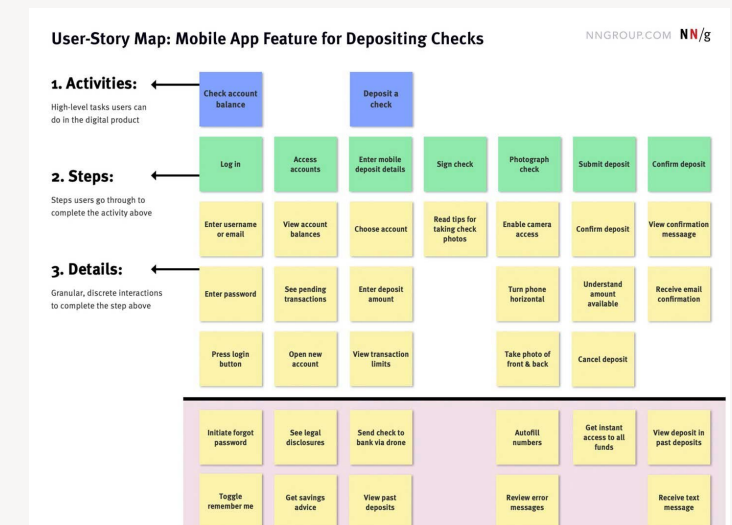
- **Collaboration tool**—a prompt for discussion.
- Useful in design work as a lightweight way to express capabilities/value.
- Can be written at different sizes: epics (big) → stories (smaller)
- ‘Epics’ are broken down into smaller stories later.
- Early on: keep them outcome/value-focused (avoid UI specifics)
- Example: *‘As a student, I would like to receive reminders of coursework deadlines, so that I remember to submit my coursework in time.’*
- See also: <https://www.interaction-design.org/literature/article/user-stories-as-a-ux-designer-i-want-to-embrace-agile-so-that-i-can-make-my-projects-user-centered>

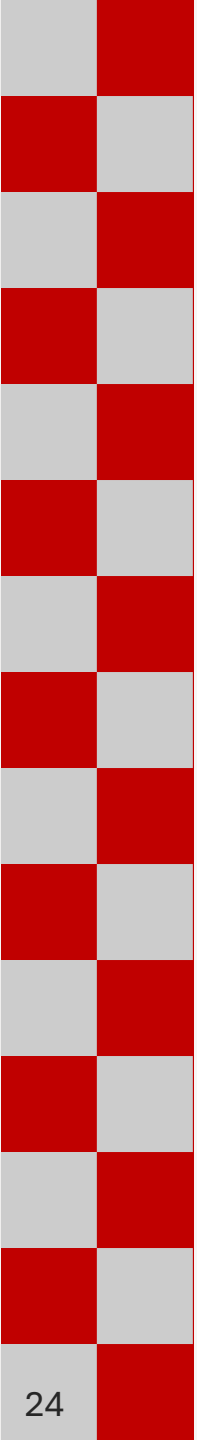
User Story mapping

- A way to organise stories into a coherent view of the product
 - Map out product features.
- Helps teams see the “big picture” and prioritise what to build first.
- Typical structure: Activities → Steps → Details
 - Activities (high-level tasks)
 - Steps (to complete the activity)
 - Details (discrete interactions to complete a step)

See also:

<https://www.nngroup.com/articles/user-story-mapping/>





Conceptual Design

What might we have after User Research?

Potentially:

- Descriptions of the current users.
 - e.g. using empathy maps.
- Descriptions of the users' 'as-is' (current) activities.
 - e.g. using current user journey maps, or task models*.
- A description of archetypal potential users.
 - e.g. using personas.
- A definition of the design problem or challenge.
 - e.g. using Point of View (POV) statements, user stories, detailed requirements, or other statements of the design goals.

*Not covered in class, but I uploaded some optional supplementary material to session 4, for interest

A reminder...

“Designers resist the temptation to jump immediately to a solution to the stated problem. Instead, they first spend time determining what the basic, fundamental (root) issue is that needs to be addressed. They don't try to search for a solution until they have determined the real problem,

and even then, instead of solving that problem, they stop to consider a wide range of potential solutions.

Only then will they finally converge upon their proposal.’”

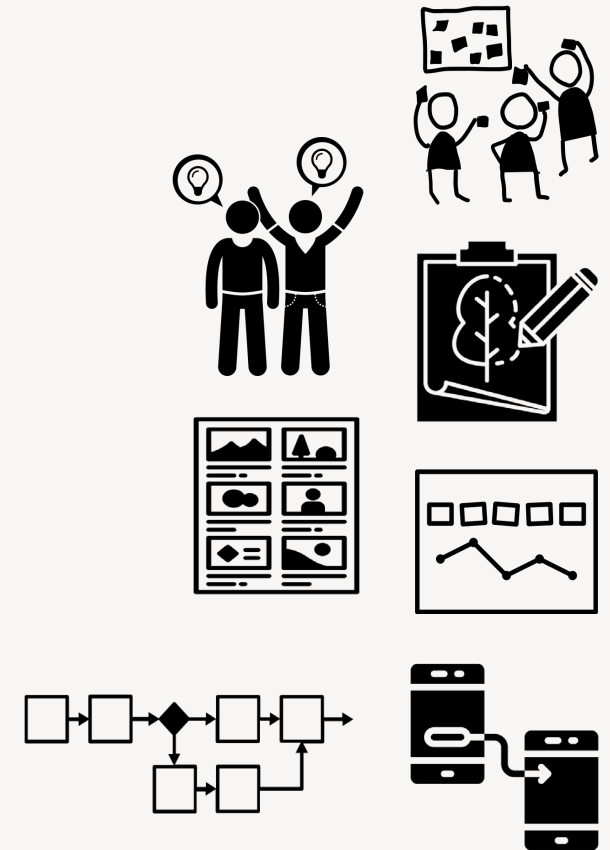
— Don Norman
Rethinking Design Thinking

Conceptual design

- Conceptual design is **high-level** design work.
- It **bridges** the *problem space* and the *design space*
- It focuses on **what kind of product or system** it is and **what facilities it must offer**, not the interface details.
- Begin by generating **lots of design ideas**.
- Ideas are **evaluated and refined**, with weaker ones discarded.
- Design is **iterative**—exploration and elimination of alternatives is repeated.
- You seldom get it right the first time—the first idea is rarely the best, so give it competition!

Conceptual design **aims** to achieve

- Generate, explore & compare alternative high-level design ideas.
- Envision users' future activities and contexts of use.
- Make early design decisions about:
 - Modes of interaction
 - Types of interface
 - Tasks and activities to support
 - Key concepts and workflows
- Gather early feedback to refine concepts.



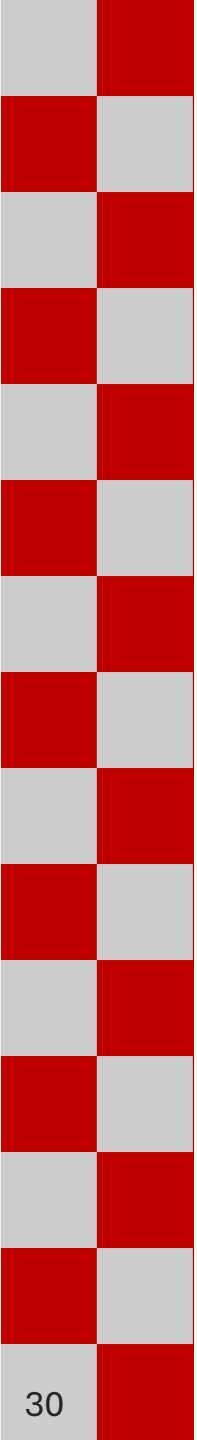
Conceptual **vs detailed** design

Conceptual Design

- High-level; focuses on what the system is and does.
- Concerned with users, goals, and context.
- *Defines: what the system will do, interaction types, interface types.*
- Not about screen layout or labels...

Detailed Design

- Screen-level; focuses on how the system will look and behave.
- Defines interface layout, navigation, and visual details.
- Specifies functional logic and interaction flow.

A vertical decorative element on the left side of the slide, consisting of a red and white checkerboard pattern.

Conceptual Design

Design Inspiration

Design **inspiration**

- Finding great ideas is a really hard part of the design process.
- Many people do one design and refine that
... because they can't think of alternatives
- You can deliberate interesting design ideas before you need them.
- Then you'll have inspiring material to draw on when you need to think of alternatives.

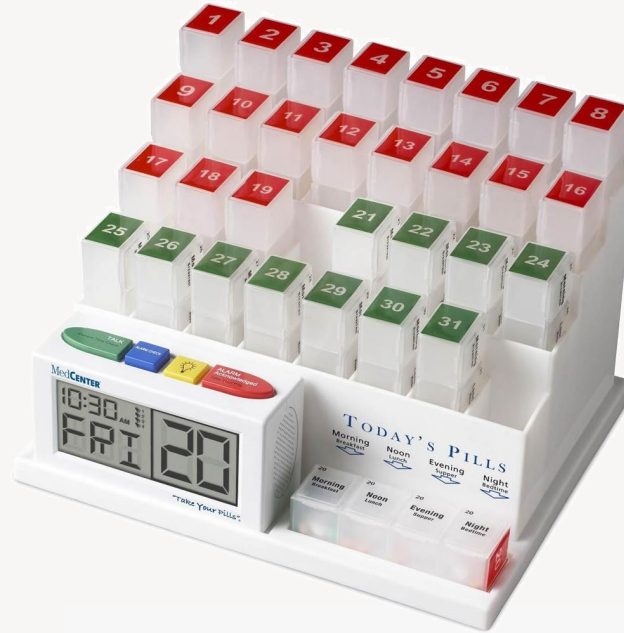
Design **inspiration** sources

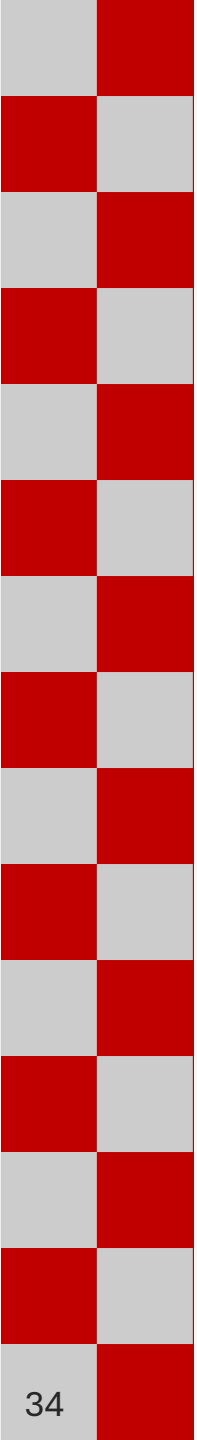
- Output from user research
- Similar projects / systems
- Different projects / systems
- Design library
- Ideation techniques
- ...



Image of textile design library
<http://www.style-diaries.com/2014/06/the-design-library.html>

e.g. Pill tracking containers





Conceptual Design Ideation

The structured, creative process of generating ideas

Conceptual design & Ideation

Two interdependent concepts

- Use early ideation to discover conceptual directions
(What could we design)
- Use conceptual design to define what the system is and *broadly* how it works
(What kind of system are we designing)
- Use later ideation to explore different ways to realise the chosen concept—help refine and elaborate it
(How might the system work in more detail?)

This process is iterative, and not strictly sequential.

Ideation

See <https://www.interaction-design.org/literature/topics/ideation>

Techniques

- Brainstorming
- SCAMPER
- Worst possible idea
- Analogical reasoning
- Constraint removal
- Crazy 8s
- And many others...

Brainstorming

- A group activity to produce lots of ideas around a given problem.
- A mix of stakeholders working collaboratively in a room with whiteboards or large table, usually with a facilitator.
- Each person notes down their ideas as they happen, and sticks them on the board / table for discussion.
- Beware straying off topic.
- Beware extroverts dominating the conversation.
- Beware introverts not being heard.
- Run a warmup activity before the brainstorm.

Adapted from

<https://www.interaction-design.org/literature/topics/brainstorming>

Elements of a **successful brainstorm**

- Pick the right environment.
- Have a facilitator.
- Set a time limit.
- Begin with a target problem/brief.
- Allow one conversation at a time.
- Refrain from judgment/criticism.
- Encourage weird and wacky ideas.
- Aim for quantity.
- Build on others' ideas.
- Stay visual.

Adapted from
<https://www.interaction-design.org/literature/topics/brainstorming>

See also: <https://www.ideo.com/pages/brainstorming>

‘How Might We’ questions

- You can **rephrase your Point-Of-View statements** to use them as a guide for discussion.

e.g. *“How might we remind Mrs Morton to take her medication?”*

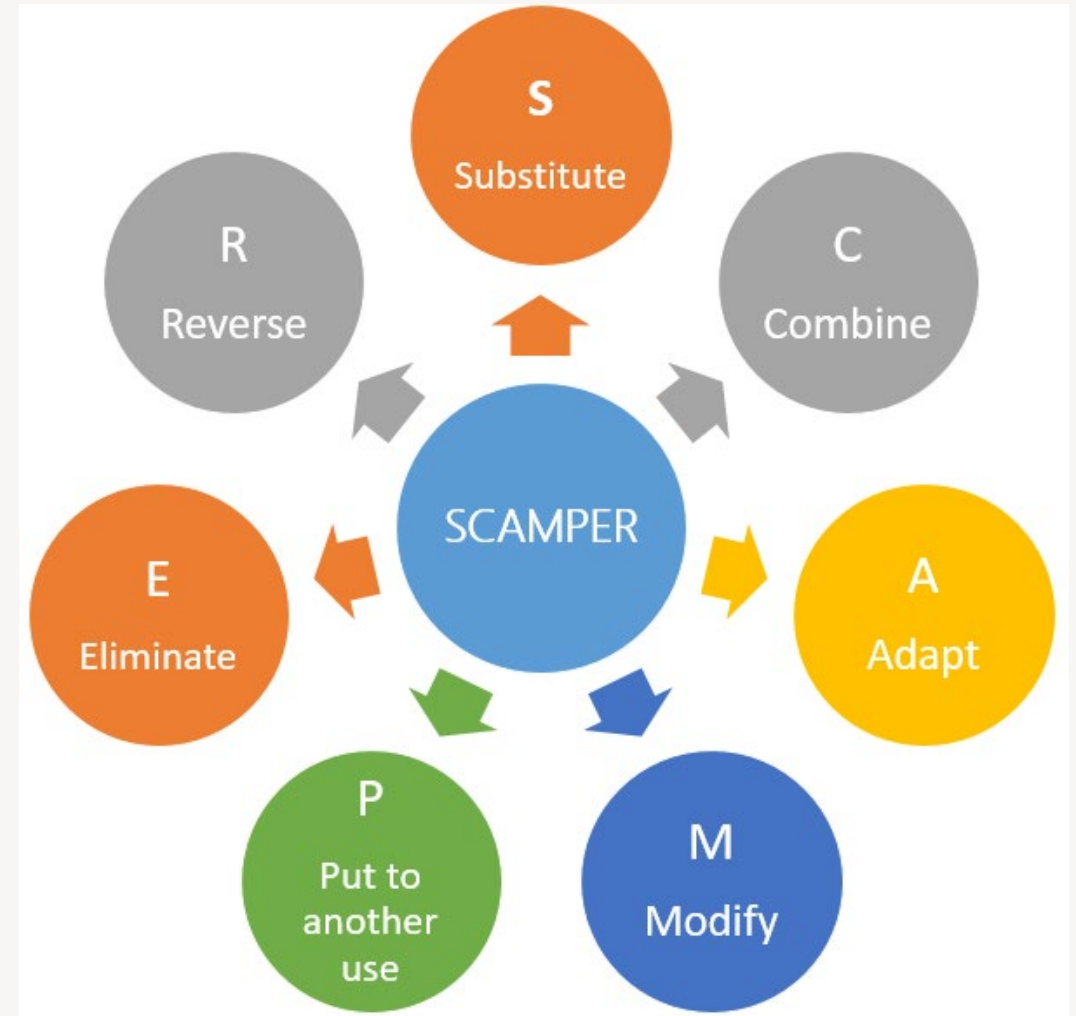
- These ‘How might we’ statements can then be used in brainstorming and other ideation sessions.
- A good way to frame challenges as opportunities for innovation.

SCAMPER

- Generate ideas for innovating on an existing product
- By looking through 7 provocation lenses:

Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse / Rearrange

- Choose your product, then ask questions for each lens



See also: <https://www.interaction-design.org/literature/article/learn-how-to-use-the-best-ideation-methods-scampers>

Constraint Removal

- Make possible what was not previously possible.
- Constraints can impose creative blocks to problem solving.
 - E.g. time, cost or budget, size....
- Think about a world without them, or where they are less important.

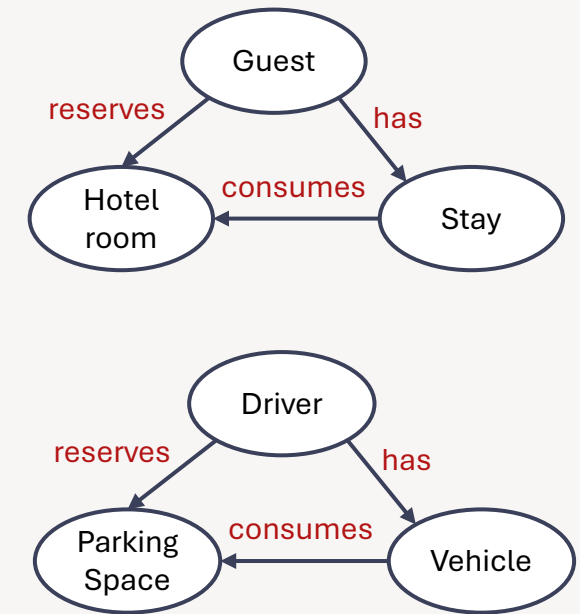


How?

1. Brainstorm to identify constraints.
2. Select constraints to remove.
3. Reduce importance of the constraint (e.g. eliminate or diminish it) to generate space for creative thinking:
 - Imagine removing one constraint at a time.
 - Generate an alternative version of the problem space.
 - Brainstorm new ideas that would be possible without the constraint.

Analogical Reasoning

- Looking to other domains for inspiration.
- Exploring solutions to similar problems in other domains.
- *'How might the solution in that domain be adapted or used as inspiration for a solution in my domain?'*
- Real-world example (Zachos & Maiden, 2008)
 - Target domain: real-time parking space booking.
 - Source domain: hotel reservation web applications.



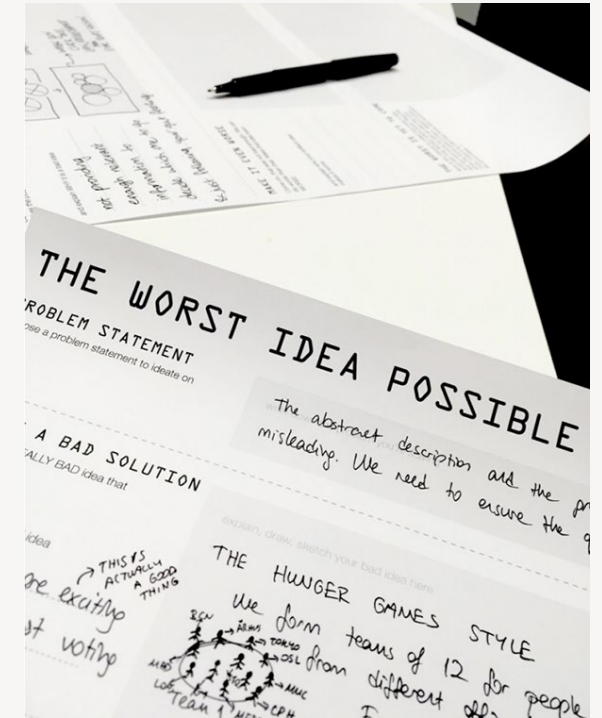
Challenge:

If designing a technology to help older people remember to take their medication, what other domains could we draw upon to potentially provide us with inspiration? Can you think of a similar problem in a different domain to use as an analogy?

Worst Possible Idea

In a group:

1. Start with a problem statement / design goal(s).
 2. Come up with as many bad ideas as possible —bad, silly, crazy, stupid, even illegal ideas.
 3. List the attributes that make them bad
 4. Then think about the *opposite* of the bad attributes or ideas, to find inspiration for *good* ideas; probe deep; what could you substitute? Mix & match?
- Removes creative pressure, no-one loses face. And it's fun!
 - See also <https://www.inc.com/bob-dorf/to-find-a-great-idea-try-looking-for-the-worst-ide.html>



From: <https://uxdesign.cc/do-a-change-and-ideate-for-bad-ideas-8b18a295e26b>

Exercise: Worst Possible Idea

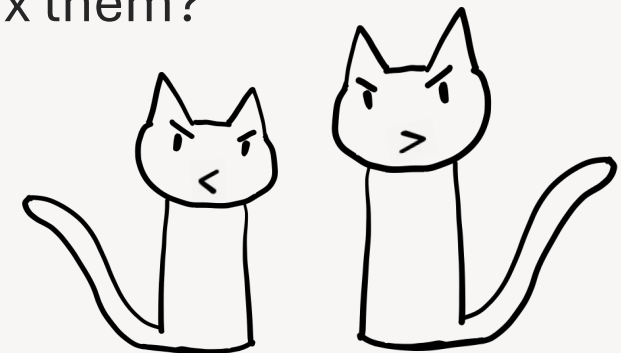
In small groups:

Design Goal: Design a digital technology to help older people remember to take their medication.

Task: Ideate, using the *Worst Possible Idea* technique

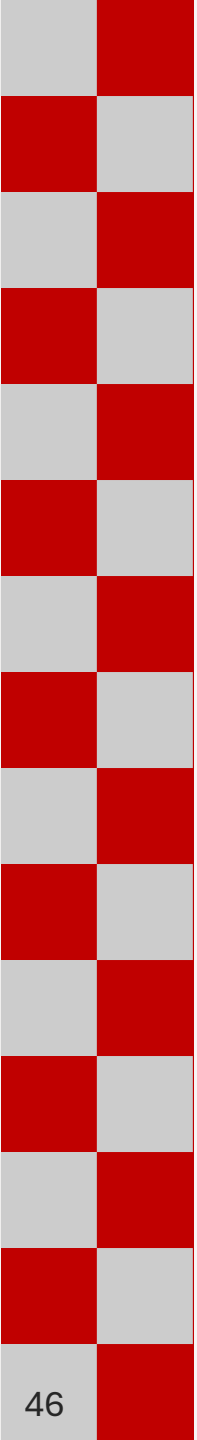
1. Come up with as many bad ideas as possible —bad, silly, crazy, stupid, even illegal ideas...
2. List the attributes (characteristics) that make them *bad*
3. Then think about the *opposite* of the bad attributes or ideas; probe deep; what about them could you substitute? Mix them?

What *good* ideas can you come up with, inspired by your *bad* ideas?



Questions?

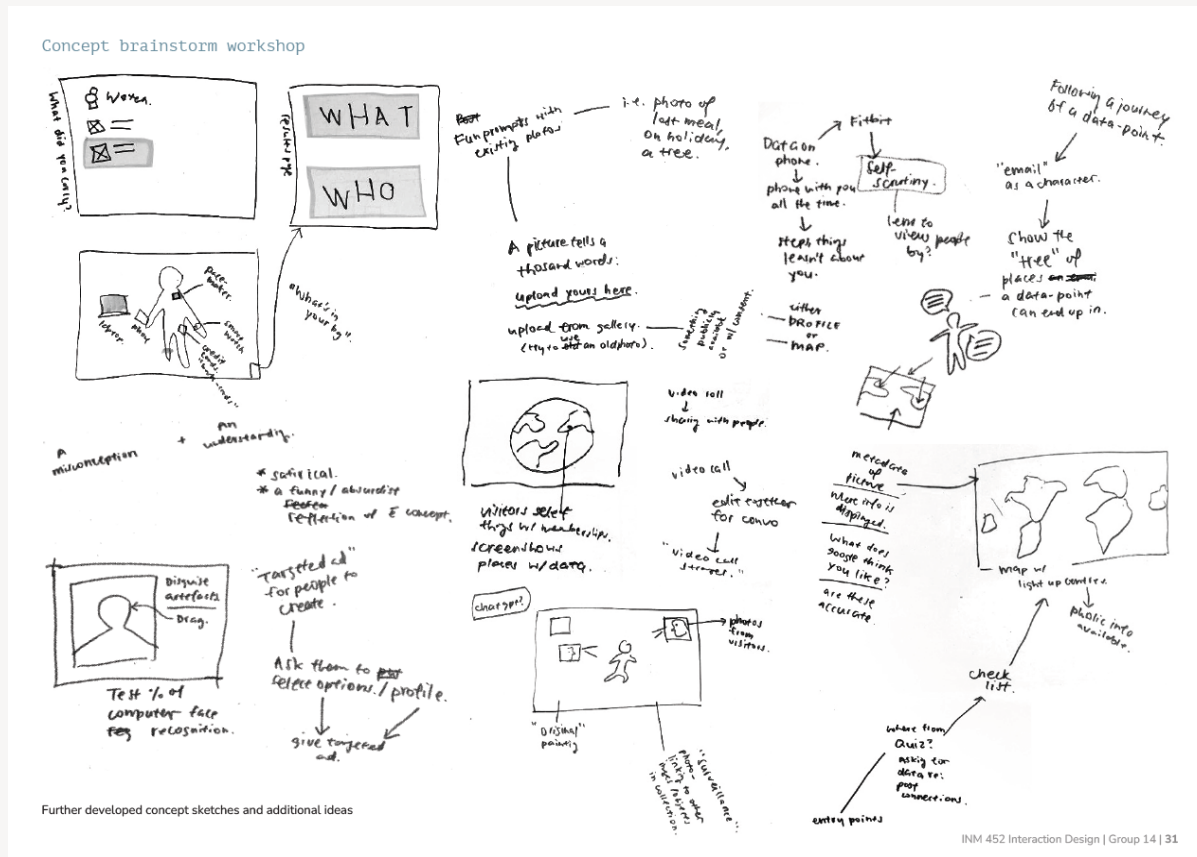




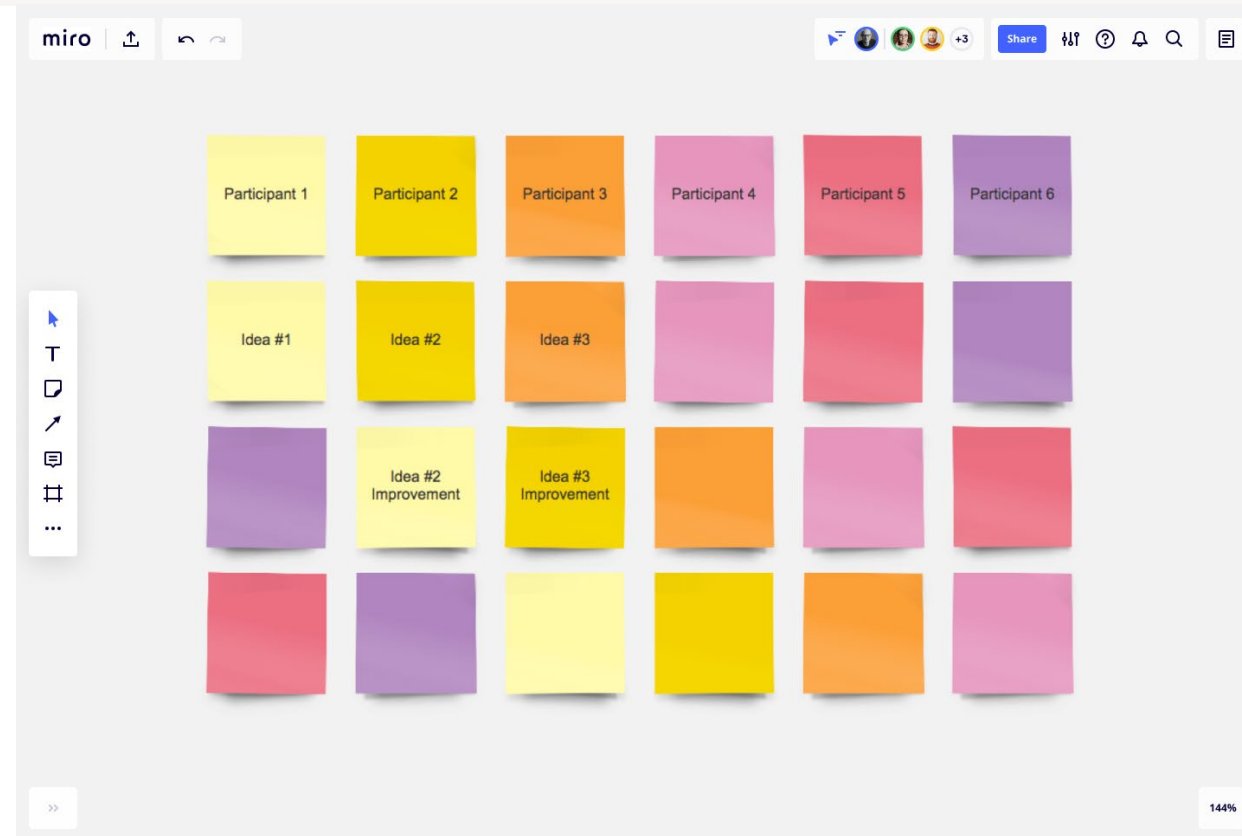
Conceptual Design

Capturing and progressing ideas

Capturing ideas



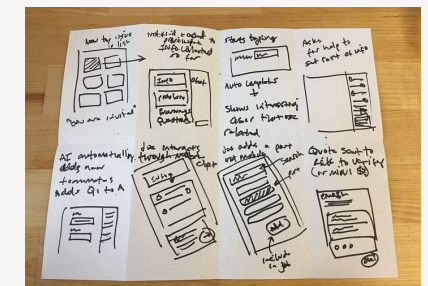
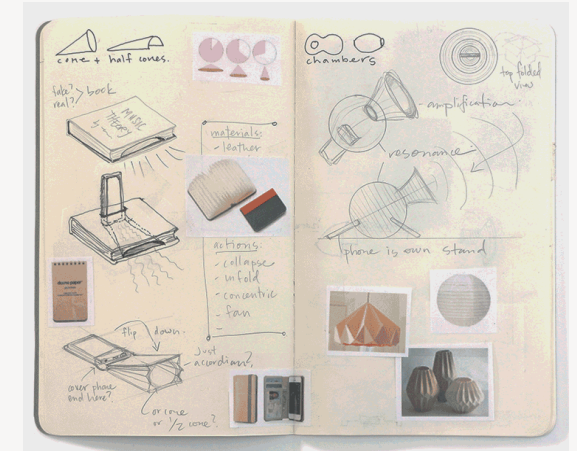
From previous student coursework

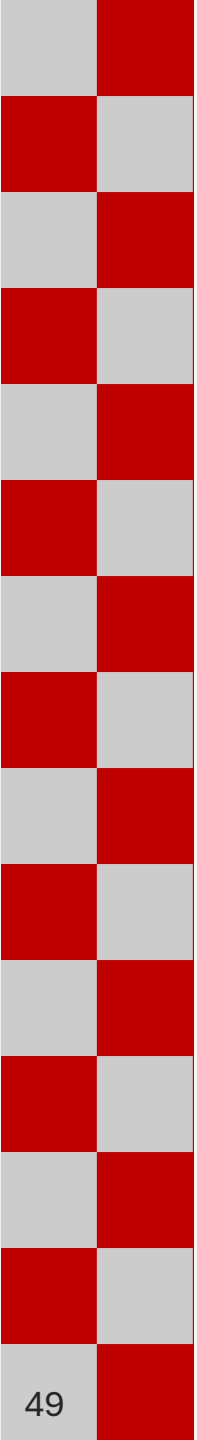


From <https://miro.com/templates/brainwriting>

Capturing and progressing ideas

- Capture your ideas in various ways:
 - Sketches.
 - Storyboards.
 - Future user journey maps.
 - Future (envisioned) tasks
 - as HTA or other task notation.
 - ...
- Progress your ideas by **elaborating**, **critiquing**, and **eliminating**.



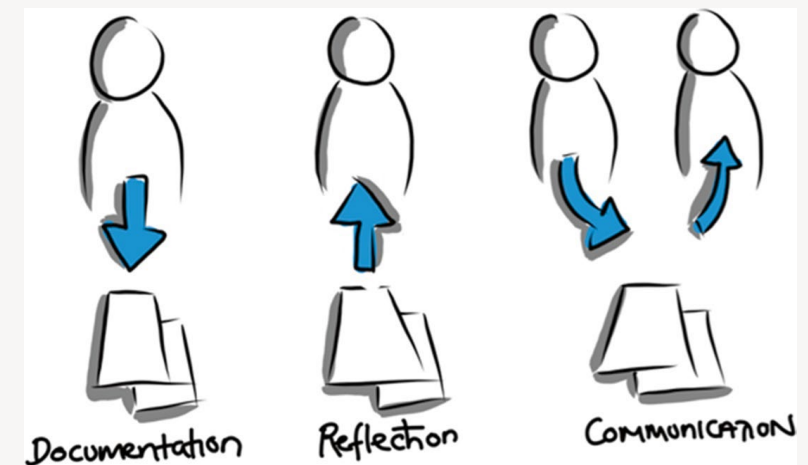
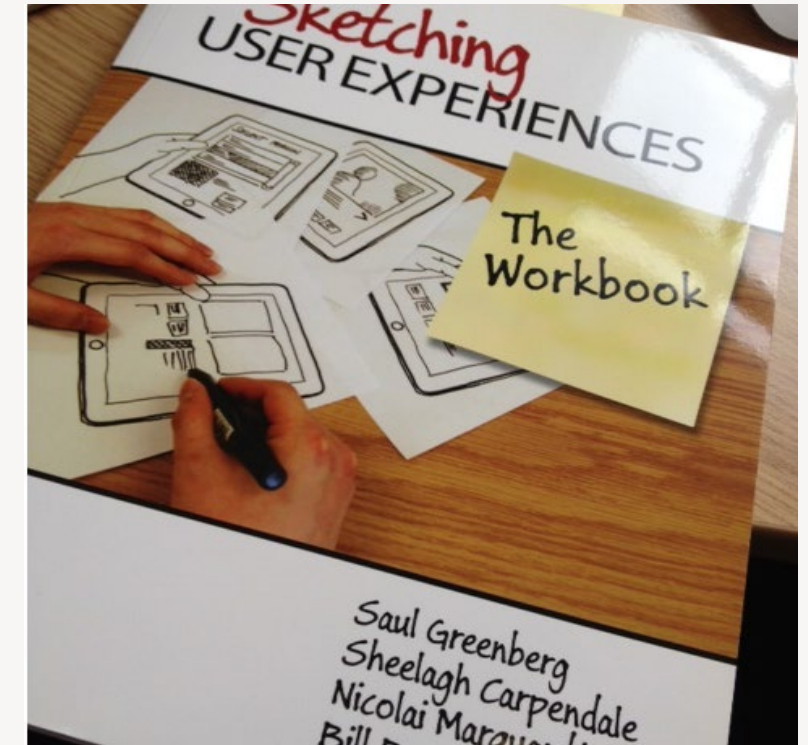
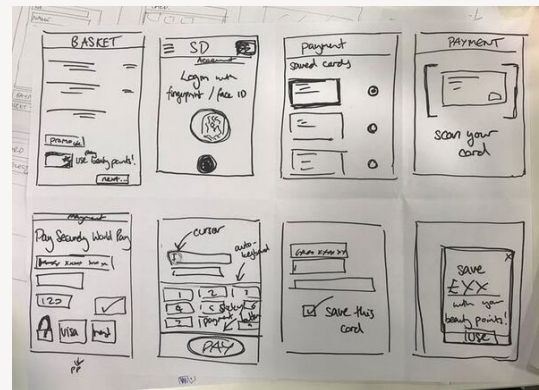
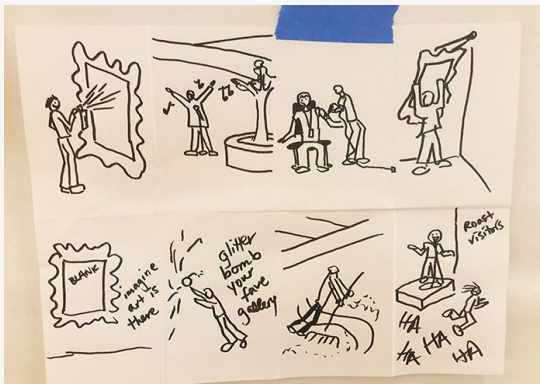


Core design techniques

Sketching

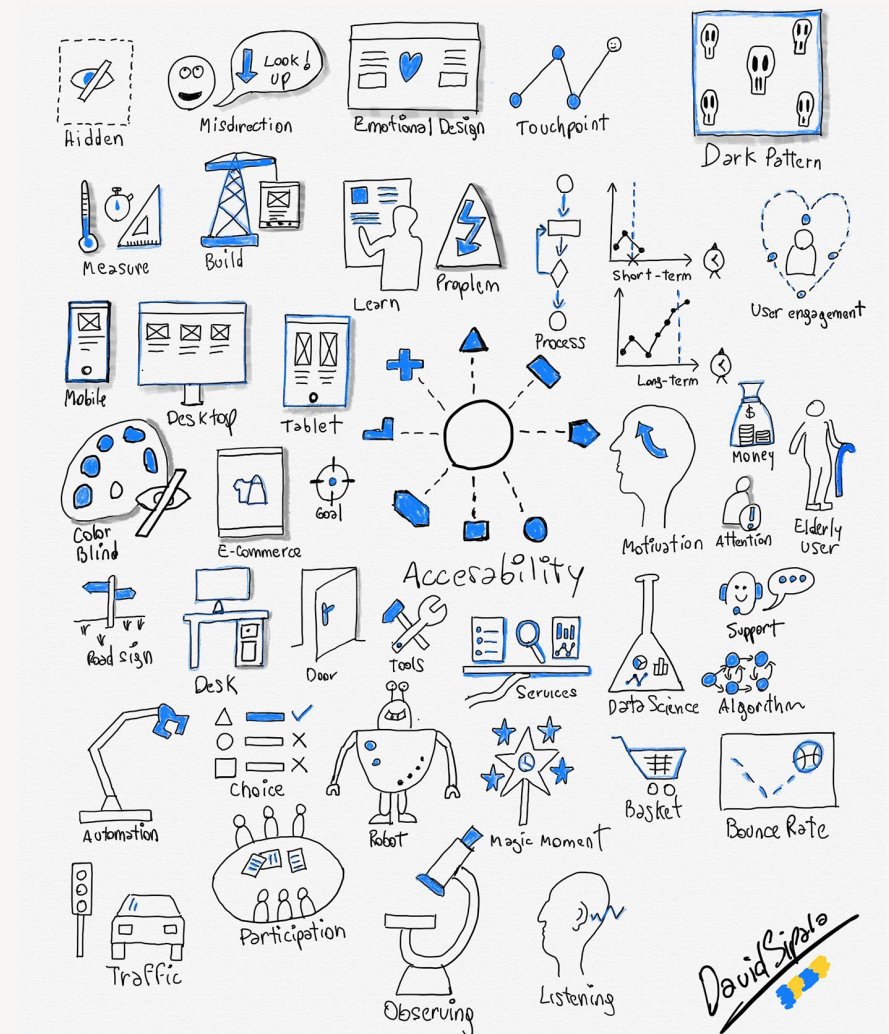
Why sketch?

- Sketching is a useful skill to **express and communicate ideas quickly**.
- Sketching allows you to **iterate** through many ideas without premature commitment.
- You can be creative and come up with innovative designs.



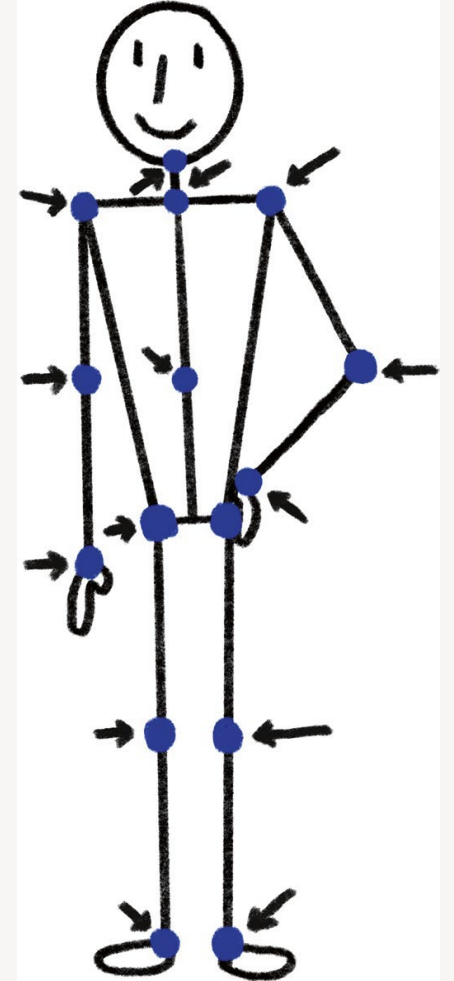
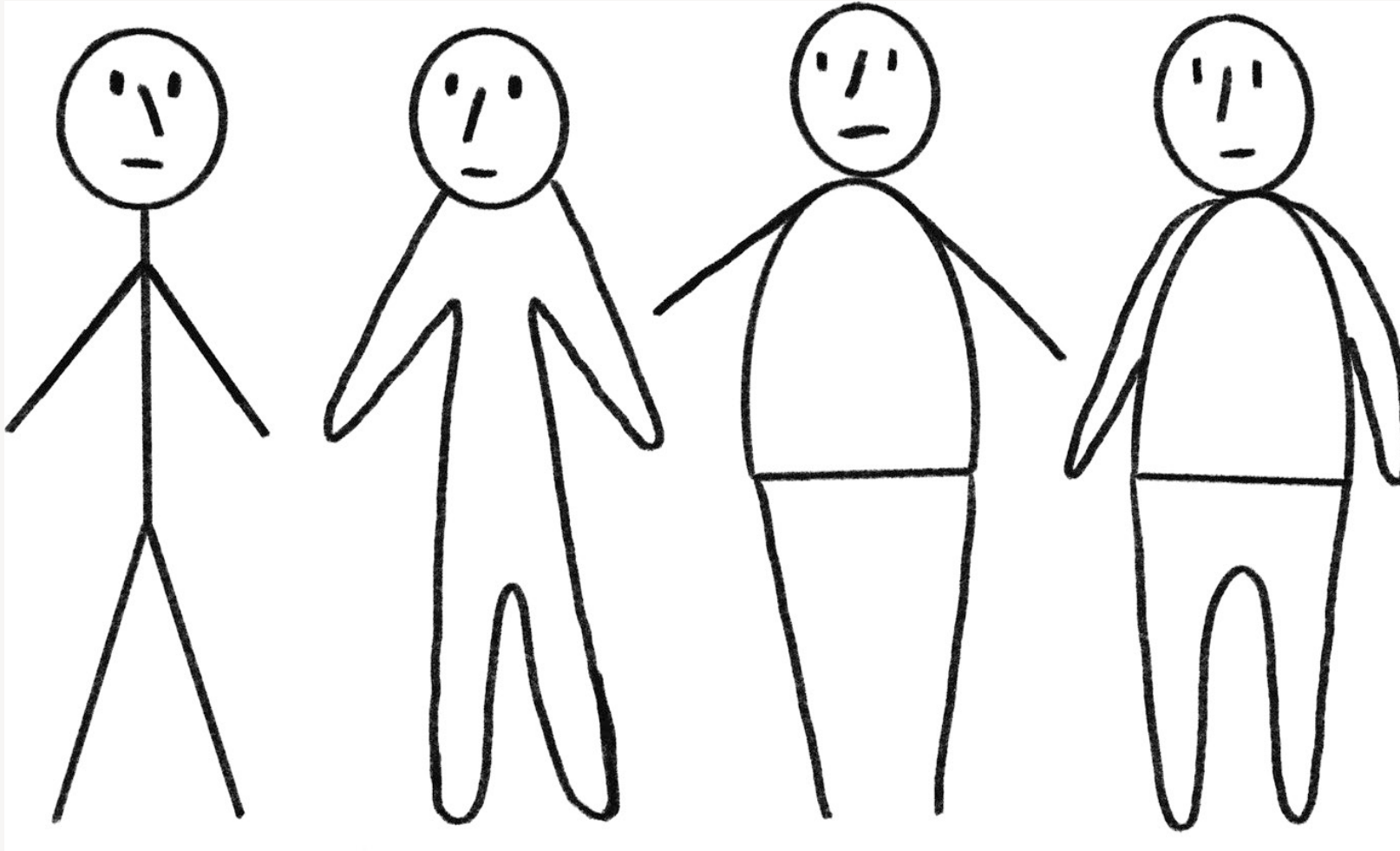
Develop a sketching **vocabulary**

- Typical digital devices
 - Mobile phone, Tablet, Laptop, Desktop computer and monitors, Mouse, Keyboard ...
- People and expressions
 - Poses, moods, emotions ...
- Abstract shapes and actions
 - Arrows, Movement, Vibration, Sound ...



David Sipala

Sketching figures



From left to right: stick person, star person, blob person, and combined result of each style. On the far right, stick figure joints. Images by Makayla Lewis, in Sketching in HCI (2024)

What do 50 million drawings look like?

Over 15 million players have contributed millions of drawings playing [Quick, Draw!](#) These doodles are a unique data set that can help developers train new neural networks, help researchers see patterns in how people around the world draw, and help artists create things we haven't begun to think of. That's why [we're open-sourcing them](#), for anyone to play with.

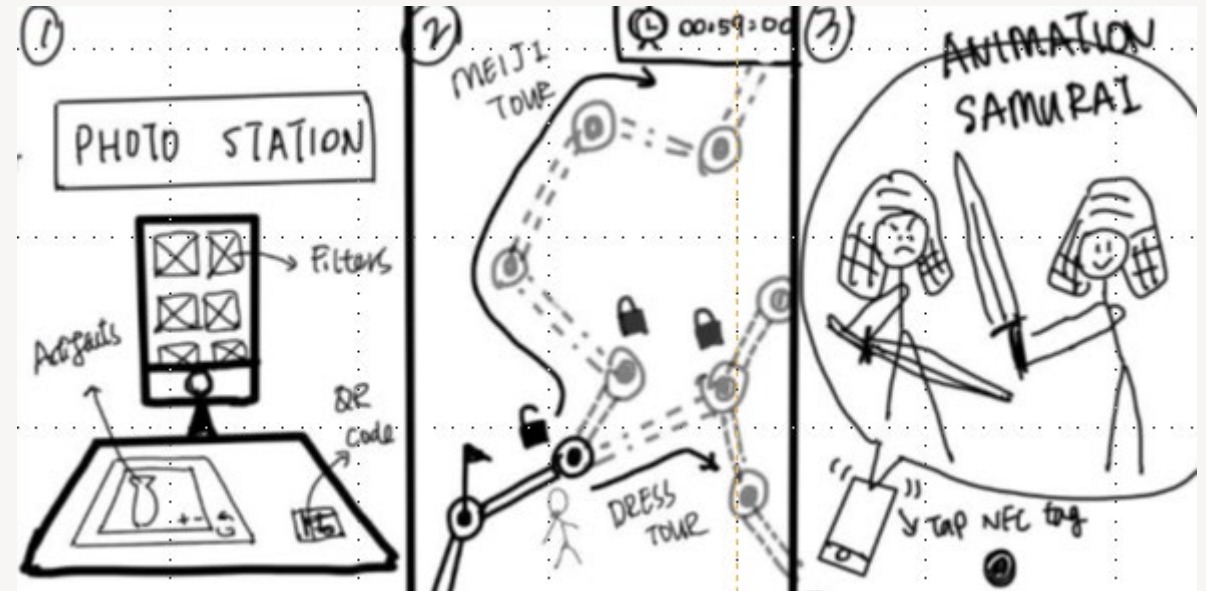
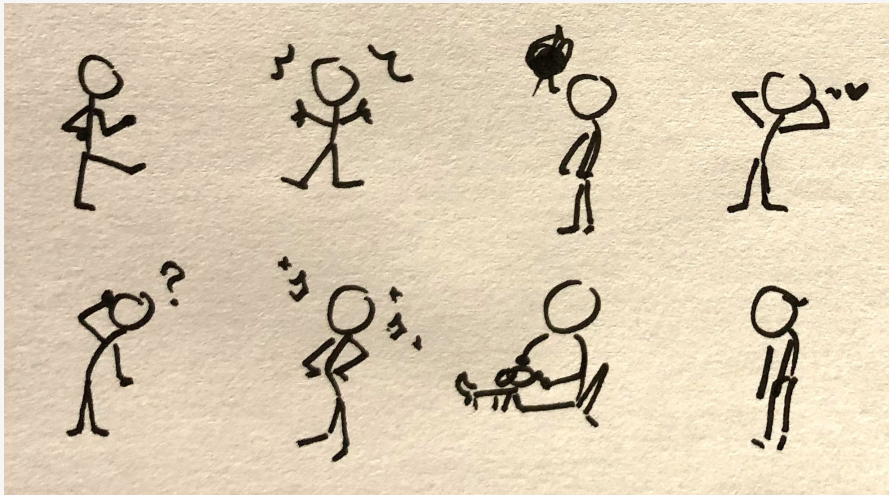
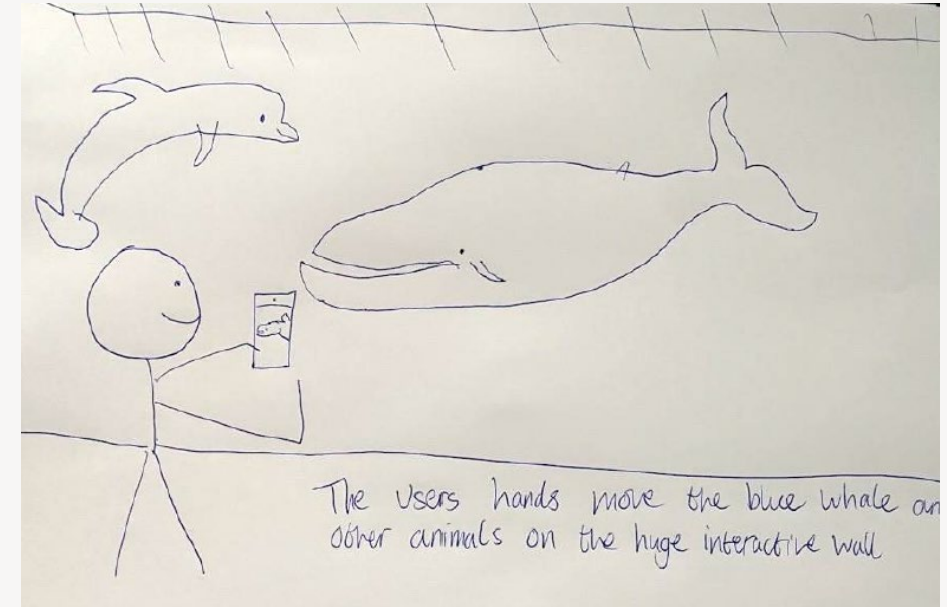
Select a drawing



<https://quickdraw.withgoogle.com/>

Sketching tips

- Faster is better.
- Be decisive!
- Don't apologise.
- Don't compare yourself.



Sketching with **stakeholders**

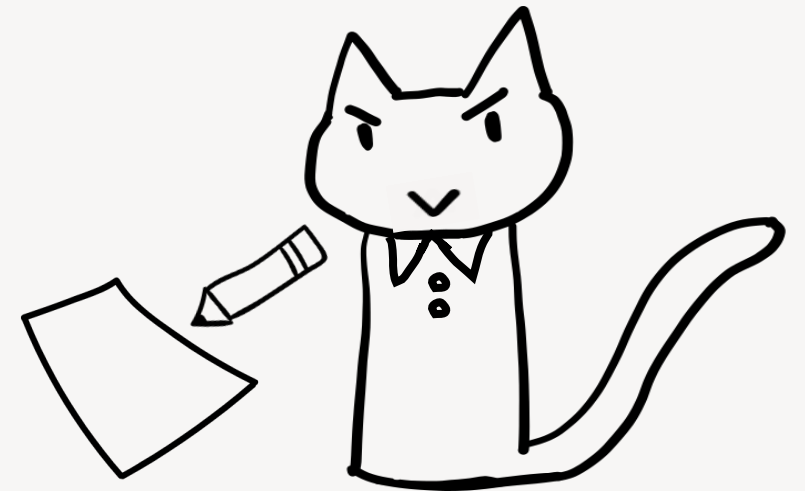
- Not everyone is comfortable with sketching
- Kaplan (2019) offers advice on how to get people outside (and maybe inside) the UX team to sketch by constraining the possibilities:
 - Fat markers
 - Tiny spaces
 - Time limits
 - Ugly examples

Kaplan K (2019) How to Get Stakeholders to Sketch: A Magic Formula

<https://www.nngroup.com/articles/how-to-get-stakeholders-to-sketch>

Exercises:

Getting to grips with sketching



fixed wrist

VISLAB.

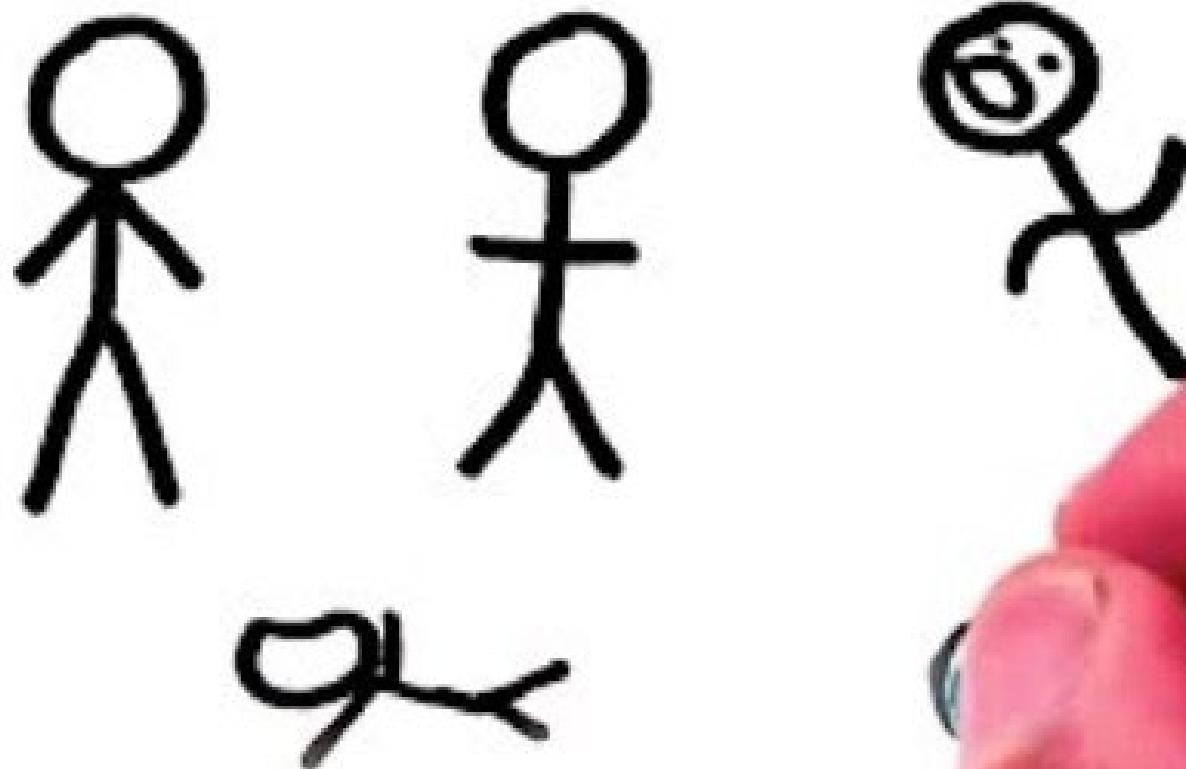


Modal Window





LAUGH • LEARN • GIVE



Exercise: **Let's sketch** something!

Individually:

1. Sketch a smartphone.
2. Sketch someone looking at a phone (whole figure)
3. Add a tree, to show that they're in a park.
4. Sketch someone looking at a map on a smartphone and being confused.
5. Sketch someone zooming into the map.
6. Sketch someone finding their current location on the map and being very happy.
7. Sketch a zoomed in version of the person, with a speech bubble saying something relevant.

Exercise: **Let's sketch** something!

Individually:

1. Sketch a ticket machine screen (just the interface).
2. Sketch the full ticket machine (screen + card slot/buttons).
3. Sketch someone using the ticket machine (whole figure).
4. Add the environment: platform sign + a clock (to show time pressure).
5. Sketch a zoomed in version of the person trying to choose a ticket and looking/feeling stressed.
6. Sketch them looking at the ticket they've bought and realising it's the wrong one, and saying something relevant.
7. Sketch them walking away angrily, to find a member of staff.

Sketching for ideation: Crazy 8s

What is it?

- A core Design Sprint method, part of the [Sketch](#) stage.
 - (usually preceded by warm-up / boot-up activities)
- Rapid sketching exercise: 8 rough sketches in 8 minutes
- A technique for generating ideas.

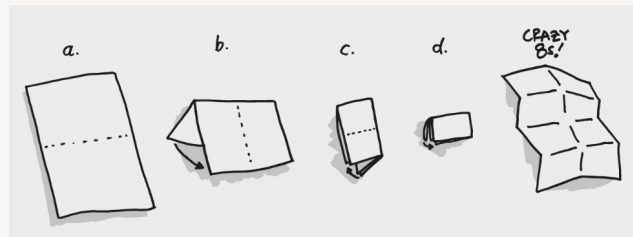
Why do it?

- Generates lots of ideas, at a rapid pace
- First ideas typically aren't the best—this technique forces you to think beyond those.
- 8 minutes (1 minute per sketch) isn't enough time for people to get bogged down in detail or perfection.
- <https://designsprintkit.withgoogle.com/methodology/phase3-sketch/crazy-8s>

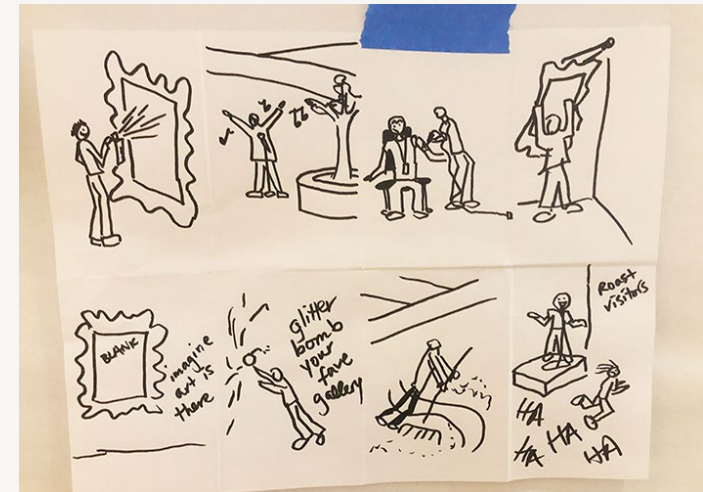
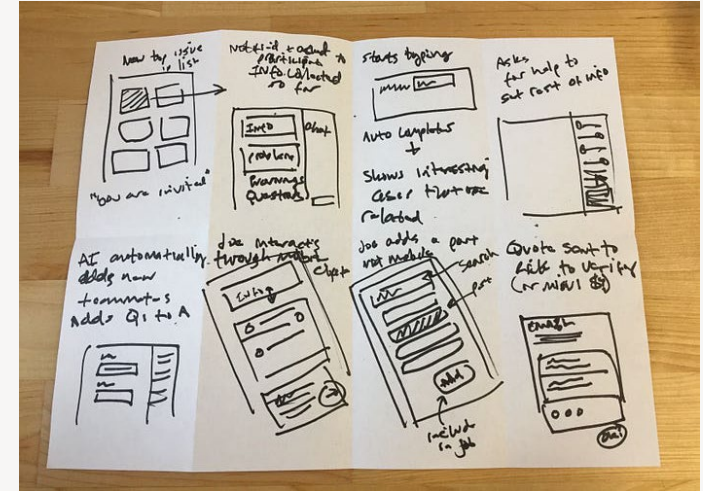


Crazy 8s: **How** to do it

1. Each person has a piece of paper with 8 squares (fold a page into 8 sections, or use a printed template)



2. Set the timer for eight minutes
3. Each person sketches one idea in each section of their paper, until they have filled all 8 sections.
4. When the timer goes off, put pens down.



Images from:

<https://uxdesign.cc/honing-crazy-eights-with-randomness-1c8100b94824>

<https://designthinkingformuseums.net/2019/07/27/bad-ideas-brainstorming>

Crazy 8s: **What next?**

- It depends on what you're trying to achieve, or the process you're following.
- Typically, ideas are first **reviewed**, to decide which to explore/develop further.
 - ↑ (more about this in the next lecture)
- When you've **shortlisted** ideas, you might try to **progress** them.
 - E.g. coming up with more detail, adapting them, envisaging scenarios for them, creating storyboards, more sketches, future user journey maps....
it depends on what you'll find useful. Experiment!
- OR you might do **another round** of Crazy 8s, using the top/shortlisted ideas as a starting point / inspiration

Questions?



Session 5 **summary**

- User research should finish with a clear definition of design goals.
 - This should capture what it is you're trying to achieve through design—the problem or challenge you are going to address, and for whom.
 - Point of view (POV) statements, user need statements, requirements, and high-level user stories are techniques you can use to define what your design work will address.
- Conceptual Design is focused on high-level design decisions.
- Ideation techniques help to generate lots of ideas to explore.
- Sketching is a core design technique and can be used to generate, explore, refine and communicate ideas



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College Building, City St.
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